

Analysis of Different College Majors and their earnings and its correlation with employment.

- How do Median Earnings differ across different college majors/degrees? And how do those earnings relate to unemployment rates? Using datasets containing different majors & major categories, employment rates, median earnings and job types for different majors the question about earnings/job prospects and their correlation to different degrees are answered in this project.

1. Problem Definition

- What you pursue a degree in while in college can impact your future when it comes to whether you are employed, how you are employed and how much you are likely to earn based on your degree.
 - To figure this out we first need to ask how median earnings differ across different college majors/degrees? And how do those earnings relate to unemployment rates?
- It is important to ask the question about median earnings for different college majors and their correlation with unemployment rates because all datasets contain a specific major and a specific category they belong in, most datasets contain the median amount of money a specific major is likely to earn and a few of them contain the data for which college degree is likely to have more employed/unemployed people in the future.
- The likely audience this project is aiming to reach which is more likely to care about the answer are students applying to colleges (these could be anyone from rising high school seniors to adults returning to college) since they're the ones who are in a position where they have to select a degree

to pursue themselves in, this will help them take into account what each degree could lead them to before finally making a decision.

2. Data Description

- This dataset was retrieved from a github repository which contained all sorts of datasets including the five csv files used in this project, all of which have something about college degree, earnings and job prospects.
- Since we have five datasets, each dataset's different rows have different meanings, breaking them down:
 - majors-list.csv: each row represents a specific major and the specific category it falls under.
 - recent-grads.csv: Each row represents a specific major and the category it falls under but goes from highest to lowest based on median earnings and other variables such as full time year round employment.
 - grad-students.csv: While each row does not specifically represent the financial condition/employment status of a specific major, it does help organize all majors and major categories in an order that'll make them less confusing.
 - women-stem.csv: Each row represents a specific major, its category and the median earnings and the total amount of people responding to the survey having pursued themselves in it.
 - all-ages.csv: Each row represents the majors and categories with specific statistical values such as P25 and P75 as well as employment.
- The key columns to research with are 'Major_category', 'Median', 'Employed', 'Unemployed' and 'Employed_full_time_year_round'. These will help with our research question since they cover the category of a student's major and their likelihood to be employed/unemployed as well as their likely median earning and their likelihood of being employed throughout the year rather than just a part time job.
- Most of the datasets contain 174 different college majors as well as the women-stem dataset containing 76 different college majors and most of them have all sorts of information about employment status, what kind of job a specific major ends up working, how much different majors end up earning and what is the employment rate for different majors.

- This dataset is missing many majors that are today a thing that once were not given that this dataset is 11-12 years old, it also could take an equal amount of people for every major since the dataset has more responses from engineering majors than majors such as Interdisciplinary and Communications & Journalism. Given all these issues we should assume that not every major will be fully represented the way they should and working with cleaning the data is one of the few ways to minimize this issue.

3. Data Cleaning & Preparation

- To clean the data, we had to classify all the different majors into a category for when we wanted to show correlations, we also had to disregard many values such as 'grad_premium' and the 'Major_code' since they had nothing to do with the question we are trying to answer about employment and earnings.
- The steps mentioned above were necessary since we needed to stop the data from being all over the place, for example there is no graph that could fit all 174 majors and their earnings to show a correlation without the x values being 'overwritten' all over one another and the chart being extremely unorganized. Disregarding irrelevant values was also important since they would take up unnecessary space that was actually needed for relevant values.
- Assumptions that were made were that Degrees that involved a lot of science and math in them had both high employment and high median earnings, degrees having a medium amount of science and math had high employment and low median salaries, and degrees not involving any math or science had low earnings and low chances of employment or even graduates working irrelevant jobs such as fast food servers (not mentioned in the dataset but can be assumed).
- Trade-offs to be considered in this dataset were that this dataset may not have data for all the majors and in fact may not take into consideration that despite having a degree in subjects such as computer science and engineering, some students may choose not to enter the tech industry directly and may be working part-time jobs until they figure out what they want to do with their degree, likewise, it does not specify where this data was collected from since different states will have different median

salaries for different professions (not in the dataset but is a fact that where you live can also impact your salary in your profession).

4. Data Understanding & Visualization

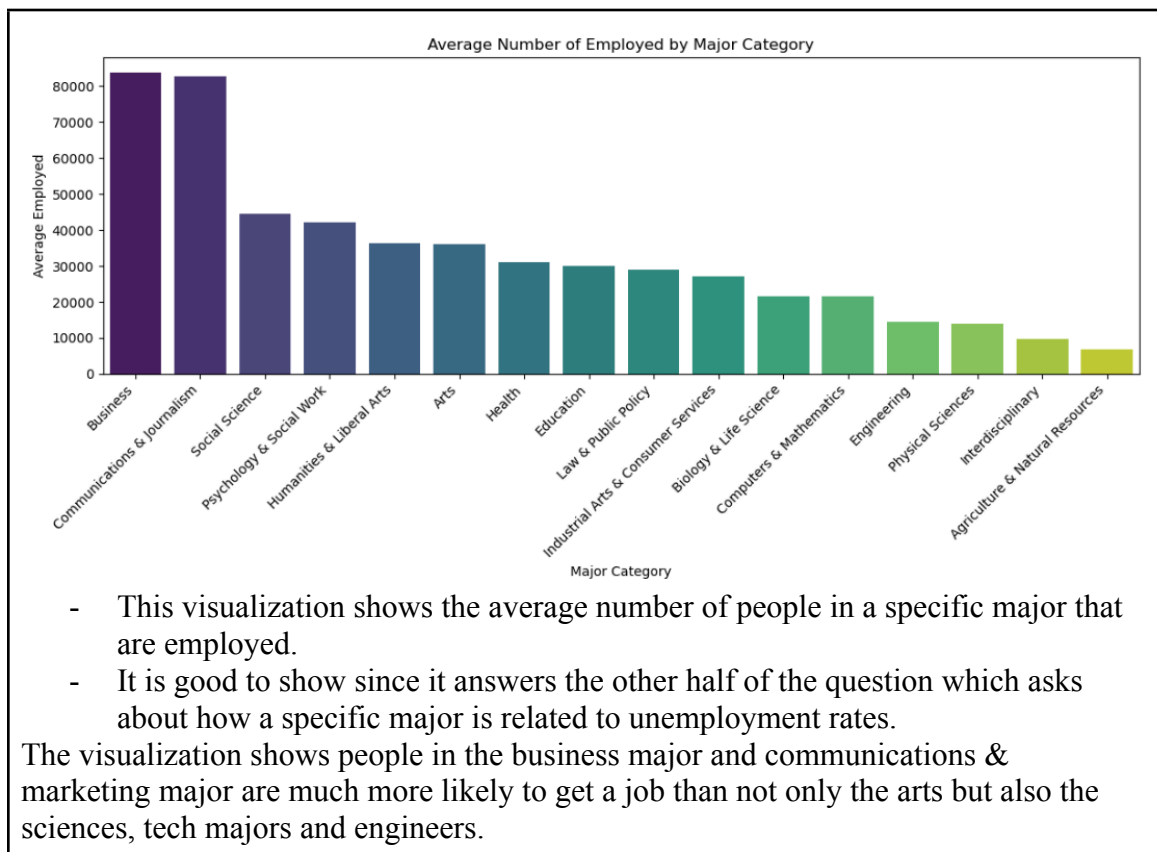
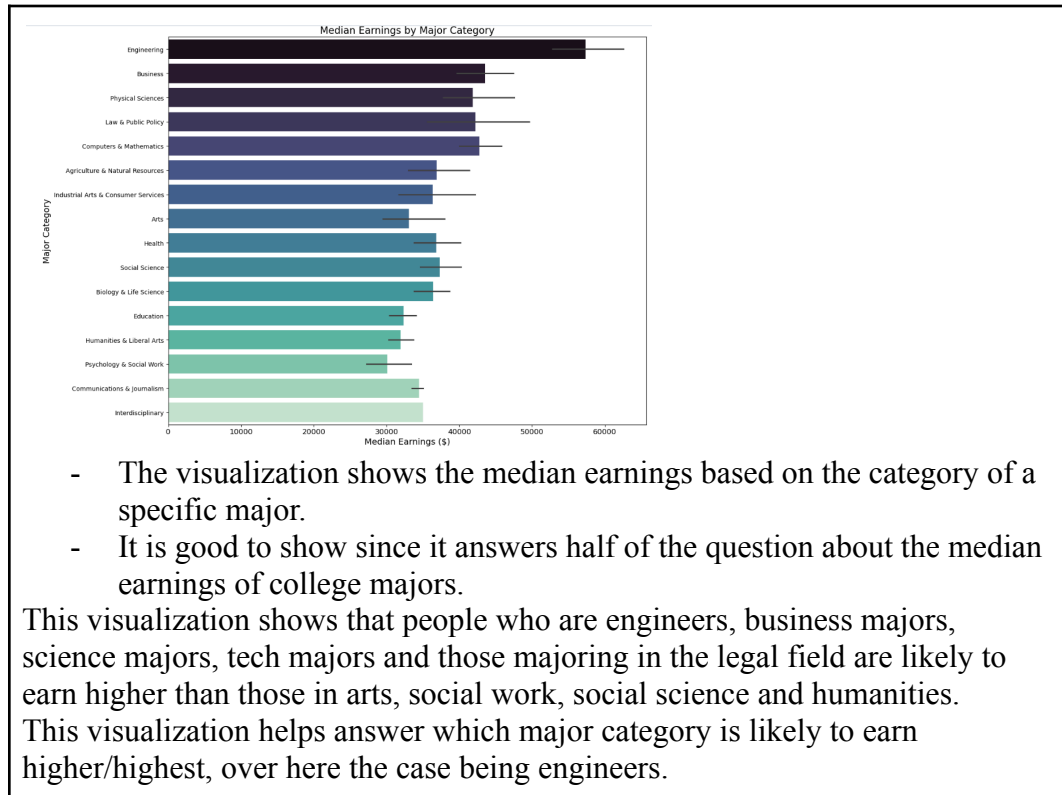
- I created visualizations that showed correlation between the job prospects, earnings and job type based on major type.

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Major_category
Engineering          29
Education            16
Humanities & Liberal Arts  15
Biology & Life Science  14
Business             13
Health              12
Computers & Mathematics  11
Agriculture & Natural Resources 10
Physical Sciences    10
Psychology & Social Work  9
Social Science       9
Arts                 8
Industrial Arts & Consumer Services 7
Law & Public Policy      5
Communications & Journalism 4
Interdisciplinary     1
Name: count, dtype: int64
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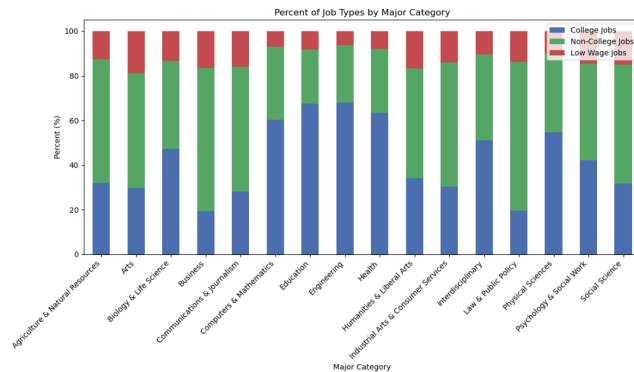
- This visualization shows the number of people in a specific major category based on everyone who responded
- This is good to show since it shows which people have more/most likely responded to the survey since it helps take into account biases that could occur.

Here it shows that Engineering was a popular major in all the respondents as well as education, at the same time there's less respondents who took social sciences, arts and interdisciplinary related majors. This shows that we are likely to have a lot of responses for engineers and teachers and therefore it'll be easy to filter out biases for them and harder for majors such as law and the arts since less respondents took them.

This visualization does not directly answer the question but helps filter out biases that will occur for sure.



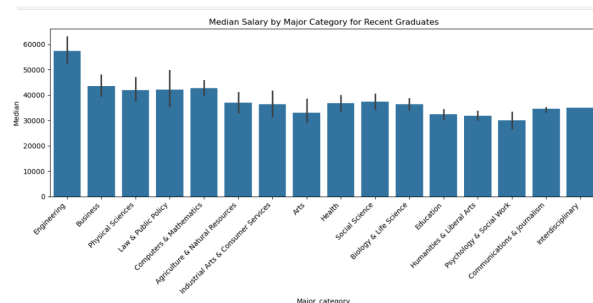
This answers the question about which major category will be more likely to get a job compared to others, which here it is clearly the communications & journalism as well as business majors.



- Even if a specific major were to get a job, what kind of job they get also matters, this visual will answer the question about what kind of jobs a specific major will get.
- It is good to show since while it does not exactly correlate to the question, it does help show that just getting a job is not the goal if it does not relate to your degree.

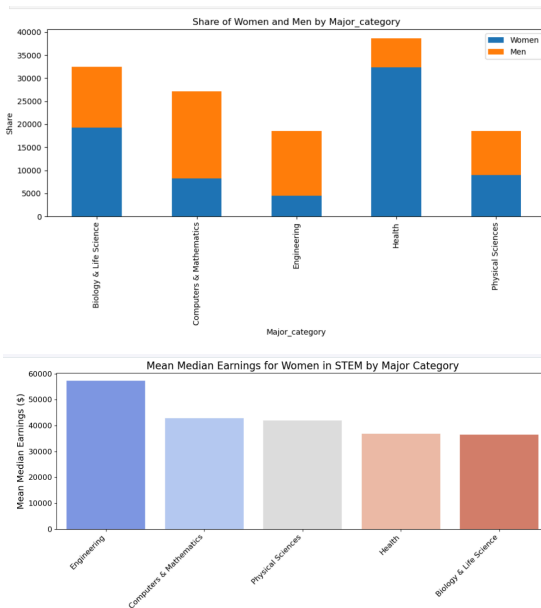
The visualization shows that technical, engineering and science related majors are less likely to work low-wage jobs, it also shows business and communication majors more likely to work non-college jobs but still earn relatively more compared to the arts and humanities majors who are more likely to work low paying jobs.

This shows that the category of a major not only impacts getting a job but also helps tell the difference between who is earning the minimum wage and who is simply earning not as much.



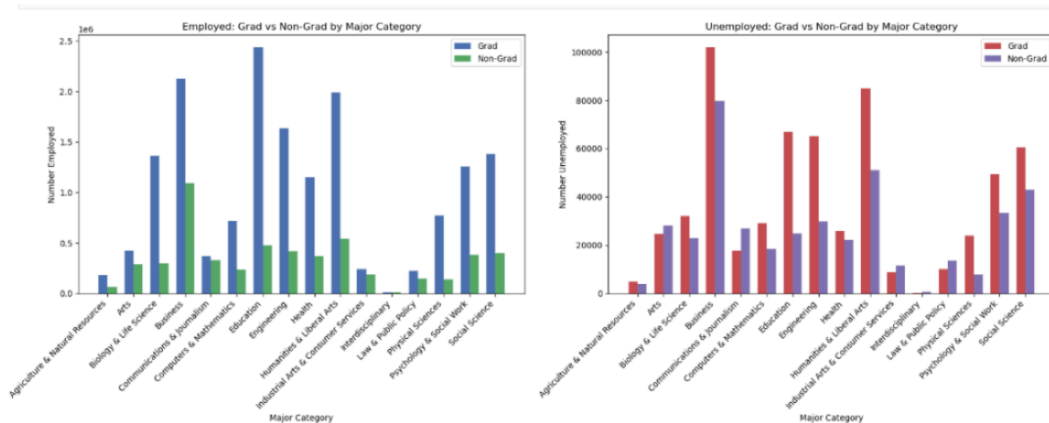
- This visualization shows specifically how much someone earns fresh out of college based on their major rather than 5-10 years after it.
- It is good to show since it also shows people that their median earnings straight

out of college and a few years after it are not necessarily going to be the same. There is not much of a difference between this visualization and the one showing earnings much later after graduation, however it does show that the category of your major will matter since your median earnings won't be changed much regardless of whether you're 22 or 32.



- The two visualizations are put together to check the correlation of the male to female ratio for a specific degree and how much that specific major is likely to earn.
- This is important since it helps answer the question about whether just the major is important to determine earnings or if there are external circumstances as well.

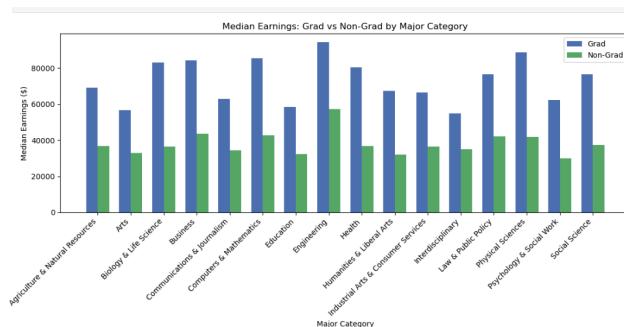
Through this dataset, we saw that engineering and tech, both male-dominated majors earned the highest; we also saw that physical sciences, health and biology & life sciences which relatively were more female dominated were paid less, if a lot of other things were not to be taken into account this dataset would show a bias. This visualization, similar to another one does not touch specifically on degree relations to major but does show that just the degree may not determine earnings since here doctors (health major category) is paid much less than engineers (engineering major category) which does have to do with the group of people in those specific majors if other circumstances are not taken into account.



- These visualizations show the likelihood of employment for a specific major based on whether they pursue themselves beyond a bachelor's degree.
- This is important since it shows which majors should pursue themselves higher than what they already are at for better job prospects.

Through this dataset, we see that most majors are equally likely to be employed/unemployed whether they pursue themselves beyond a bachelor's or not, however, those encouraged to pursue themselves higher than a bachelor's to get a job would be arts majors, communication & journalism majors, people majoring in industrial arts & consumer services, people in interdisciplinary majors as well as those in law & public policy.

This visualization shows that specific majors won't just get you a job but also increase the likelihood based on whether you've pursued yourself beyond just a bachelor's in that specific field, this shows the importance of having a graduate degree even in a subject that has no science or math in it, a perfect example here is the arts which will more likely get you a job if you have a master's than just a bachelor's.

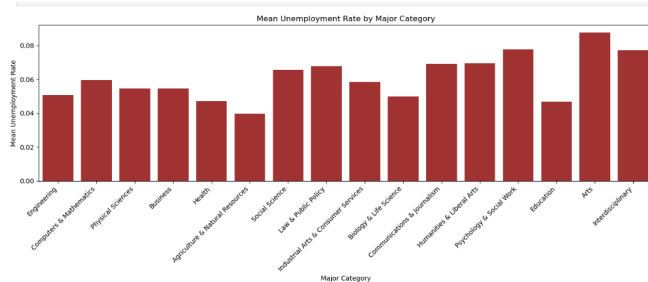


- This visualization shows how much a specific major will earn based on whether they pursue themselves beyond a bachelor's degree.
- This is important since it shows which majors truly have an advantage if they pursued higher than a bachelor's degree.

This is a pretty simple visualization since it shows that regardless of major, pursuing higher than a bachelor's degree will increase your median earning.

This visualization will help those students who want to earn more but don't know if a

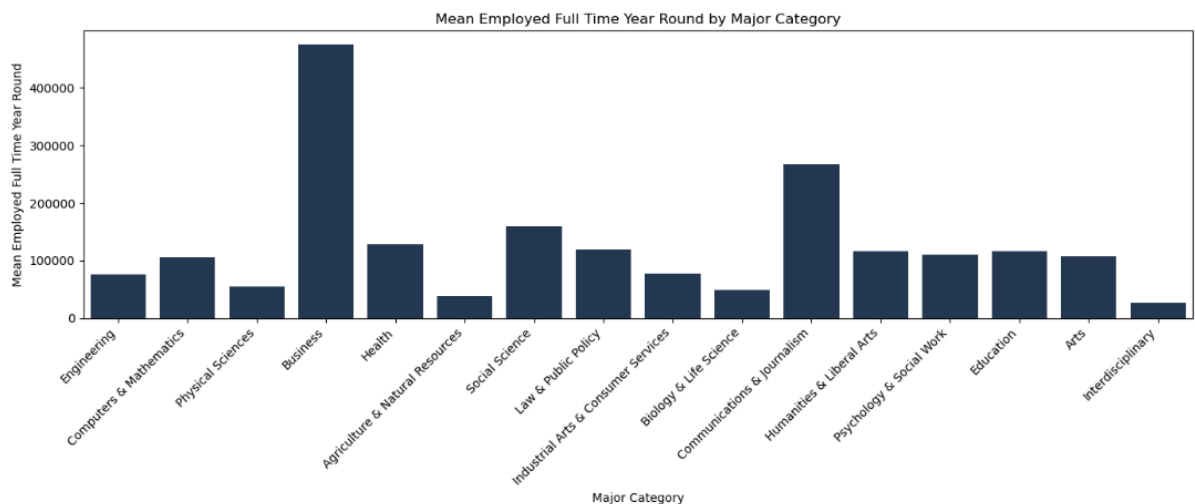
master's/Phd in their field will really help much.



- This visualization shows the average unemployment rate based on the category of a specific major.
- Given that we already have a visualization on correlation between major categories and employment, what's the need for this? Here comes into play the fact that all different major categories had a different number of responses which was going to cause some kind of bias skewing left or right, here we are actually dividing the part by the whole which eliminates the previous bias.

This visualization shows very similar results to the previous visual where non-technical majors are more likely to be unemployed than technical majors who are in turn more likely to be unemployed than majors such as agriculture & natural resources.

This answers the question about which kinds of majors are more likely to be unemployed taking into account the amount of respondents there were in the dataset for each major, therefore minimizing the bias.



- This visualization shows the average amount of people having a full time job throughout the year based on the major.
- This shows which majors are likely to not only be employed, not only having a well

paying job but also have it for a long time.

This visualization shows business being a degree guaranteeing a high paying job, a guaranteed job and a guaranteed long term job. It also shows that technical majors are guaranteed more pay but less guarantee of a job or a full-time one, as well as arts/humanities majors having relatively less likelihood of employment, less pay and less likely a long term job.

This answers the question about which majors can not only expect good pay and a guaranteed job but also which majors are likely to have it full time and keep it for the long term, over here business clearly showing that result.

5. Storytelling & Interpretation

- All these visuals and the conclusions made from them directly connect to the question about ‘How median earnings differ across different college majors/degrees? And how do those earnings relate to unemployment rates?’ through:
 - Showing how likely a specific person is to get a job based on their major which are: Business, Communications & Journalism and the Social Science majors.
 - Showing which kind of jobs someone is likely to get based on their major, which in this case has business and communication majors getting jobs that aren’t related to college but still pay well, arts & humanities majors more likely to get low paying jobs and technical majors more likely to get college related jobs.
 - Showing which majors are likely to earn more based on their major, in this case business, engineering, technical science and law majors are likely to earn more than majors in social work, humanities and arts (this is applicable also to those fresh out of college than just 5-10 years after their bachelor’s).
 - Showing which majors should pursue themselves beyond just a Bachelor’s degree in order to earn more and be more likely to land a job, in this case:
 1. To earn more people pursuing a major in the category of all the majors that responded to the survey should pursue themselves in more than just their bachelor’s degree.
 2. To be more likely to land a job, people pursuing a major in the category of non-technical, non-science related and non-engineering majors should pursue themselves beyond a bachelor’s to increase their likelihood of landing a job.

- Overall, people that want to have relatively less trouble finding a job, want to mostly earn more money compared to other majors and find a guaranteed job for the long term and many more advantages and benefits should major in business, all the visualizations show business majors having an advantage to most other majors regardless of category, which would mean someone pursuing a degree higher than a bachelor's in business has the best advantage out of all the majors. When it comes to someone whose top priority is a high paying job not taking into account job prospects or long term stability as much they should select majors more on the technical side such as engineering, technical majors and the sciences as well as law & public policy, these majors are encouraged to pursue beyond a bachelor's degree to increase chances of both pay and getting the job/ When it comes to someone whose top priority is finding and keeping a job rather than earning relatively more money, people in the category of majors such as agriculture & natural resources, communications & journalism and education are more likely to fit these criteria and are encouraged to pursue these specific majors.
 - Business majors have an advantage compared to everyone else/most of the other majors in all cases, however, based on whether a specific person chooses to pursue themselves beyond a bachelor's in a specific major can also put them at an advantage in the end.
- Something that could be misleading to the audience is those that are considering dual majoring or even adding a minor since they have not added that in the dataset, and therefore only doing a business major will not guarantee a good and well paying job compared to others or only doing engineering will not guarantee a well paying job compared to others or only doing communications & journalism will not guarantee a good job compared to others.

6. Limitations, Ethics and Overall Reflection

- The data does not capture many majors that are a thing such as but definitely not limited to: Bioinformatics, Artificial Intelligence, and Cybersecurity which are majors/degree programs at UNC Charlotte alone. Even if all majors from every US university were taken into account, it would be extremely difficult to organize them all into a visual since they all would take up all the space and create very unorganized scatterplots, barcharts, etc. due to the fact that they'd all be cluttered together, which is why they had to be put into a major category which while

helpful does not go deeper into the specific major, for example: ‘astrophysics’ and ‘oceanography’ are both in the category of ‘physical sciences’ but are completely different things and therefore would confuse any student wanting to select a specific major.

- Some biases that definitely exist is the lack of many of the majors, some majors having more respondents than others and the dataset itself being 11-12 years old, this is important since it’s outdated in many ways since many majors/degrees did not exist fully back then or were not offered at most institutions, as well as taking into account the advancement of AI in the last 3.5 years which has taken a toll on the scope of different degrees, something the dataset can’t help with because it’s old.
- The assumption that a specific major is correlated with median earnings, which in turn is correlated with job prospects could drive everyone/most students to the same major which is not only bad for the university but could cause saturation in the job market for that specific major.
- Something that was surprising was that health majors(doctors) were likely to earn less than engineers which would have seemed normal if not for the huge difference in their median earnings, I was also surprised to see a communications & journalism major have a higher chance of getting a job than education majors or law majors which again would be normal if not for their huge difference of likelihood to get a job.
- If there was more time or data, it would be interesting to see the information for double minors, double majors, the GPA of those that had responded, as well as the degree level the grad students completed (Master’s or Phd) since that would limit the bias that was present this time and also take into account more than just the bachelor’s degree or how many people were in a specific field.

7.Code

- https://github.com/Ojas2011/College-Degrees-and-their-Multiple-Outcomes/blob/9b0c440f51340b6ca96134a97bdfea56b2e7e9ea/College_Majors_and_their_outcomes.ipynb

8. References & AI Use Transparency

- <https://github.com/fivethirtyeight/data/tree/4c1ff5e3aef1816ae04af63218015066e186c147/college-majors>
 - The Above Link is where I got all the csv files used in this project as well as the readme and the Rscript.
- For a lot of the code, previous coding assignments such as 'Pandas Demo' were used for reference with the specific code to write, for example the pandas demo assignment from January 28th provided me with the code to help 'read' a '.csv' file.
- As for AI Use, Github Copilot (which is embedded in Visual Studio Code) was used to create all the visualizations (by providing the code needed for the specific asked visualizations, be it scatterplots or stacked bar charts).
 - AI was also used in cases where code was faulty, this could include something as simple as spelling errors or capitalizations.